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Effectiveness of equipping bus stop shelters with cooling and filtering systems in a city with tropical climate

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Purpose

Waiting for a bus may represent a period of intense exposure to traffic particles in hot and noisy conditions in the street. To lessen the particle load and tackle heat in bus stops a shelter was equipped with an electrostatic precipitator and a three-step adiabatic cooling system capable of dynamically adjust its operation according to actual conditions. This study evaluates the effectiveness of the Airbitat Oasis Smart Bus Stop, as the shelter was called, to provide clean and cool air.

Study design/methodology/approach

The particle exposure experienced in this innovative shelter was contrasted with that experienced in a conventional shelter located right next to it. Mass concentrations of fine particles and black carbon, and particle number concentration (as a proxy of ultrafine particles) were simultaneously measured in both shelters. Air temperature, relative humidity, and noise level were also measured.

Findings

The new shelter did not perform as expected. It only slightly reduced the abundance of fine particles (-6.5%), but not of ultrafine particles and black carbon. Similarly, it reduced air temperature (-1°C), but increased relative humidity (3%). Its operation did not generate additional noise.

Practical implications

The shelter's poor performance was presumably due to design flaws induced by a lack of knowledge on traffic particles and fluid dynamics in urban environments. This is an example where harnessing technology without understanding the problem to solve does not work.

Originality/value

It is uncommon to come across case studies like this one in which the performance and effectiveness of urban infrastructure can be assessed under real-life service settings.

Keywords: Bus stop; Bus shelter; Public transport; Traffic particles; Particle exposure

Introduction

For urban dwellers, the greatest exposure to air pollution generally occurs while commuting or doing activities in close proximity to emission sources, such as walking along busy roads or while standing at bus stops (e.g. Poom *et al*., 2021; Shairsingh *et al*., 2018; Borge *et al*., 2016; Moosavi *et al*., 2015). Several studies have shown that microenvironments affected by vehicular traffic are hotspots of air pollution, and in turn of personal exposure (e.g. Velasco *et al*., 2019; Li *et al*., 2017; Morales-Betancourt *et al*., 2017; Tan *et al*., 2017; Kaur and Nieuwenhuijsen, 2009). Thus, commuters using public transport are likely to experience high exposure to air pollution during daily commutes (Cepeda *et al*., 2017; de Nazelle *et al*., 2017; Knibbs *et al*., 2011).

Bus stops are among the most critical microenvironments affected by traffic emissions. They are usually at curbsides, and often near to intersections and traffic lights. This puts commuters in close proximity to fresh fumes rich in particles and noxious gases emitted by passing, idling and accelerating buses and cars. Waiting for a bus may represent a period of disproportionately high exposure to air pollution. Studies have shown that commuters may experience concentrations of harmful particles several folds higher than those at ambient level, including short-lived spikes one order of magnitude higher as a direct consequence of accelerating buses (e.g. Velasco and Tan, 2016; Moore *et al*., 2012; Wang *et al*., 2011; Hess *et al*., 2010; Dales *et al*., 2007). A previous study in Singapore reported concentrations of fine particles (≤2.5 µm in size, $PM_{2.5}$) in bus stops 1.5-3 times above mean ambient concentrations reported by the local air quality monitoring network (Velasco and Tan, 2016). In terms of ultrafine particles (≤100 nm in size, UFP), the same study reported mean concentrations 2-7 times higher in comparison to those observed in a site not affected by vehicular emissions. It was found that commuters could expect at least one $PM_{2.5}$ outburst over 100 µg m^{-3} with concentrations of UFP above 100×10^3 # cm^{-3} every 10 min. However, the main concern was probably the large number of particles in nucleation mode (<50 nm) to which commuters were exposed. The main bulk of particles exhibited a size of 15-40 nm.

Nucleation mode particles are linked to combustion and non-combustion (e.g. engine braking conditions) processes into the engine and tailpipe, and condensation and nucleation mechanisms after the hot exhaust gases have passed through the emission control devices (Rönkkö and Timonen, 2019; Morawska *et al*., 2008). Because of their small size they exert higher toxicity than larger particles and are efficiently deposited by diffusion mechanisms in all regions of the respiratory tract, which lead to the development and exacerbation of respiratory illnesses via oxidative stress and inflammation (Leikauf *et al*., 2020). These particles can easily reach the alveoli (hollow, distensible cup-shaped cavities in the lungs where pulmonary gas exchange takes place), translocate into the systematic circulation, and diffuse into all organs including the brain, liver, spleen, bone marrow, and heart (Oberdörster *et al*. 2005). Furthermore, growing evidence suggests that short term exposure to UFP increases arterial blood pressure, the most important risk factor for cardiovascular disease (Ohlwein *et al*., 2019).

For the case of Singapore, standing at a bus stop for 10 min may sound innocuous. However, a commuter who takes a two-way trip for five days per week is actually exposed to traffic particles for nearly seven hours per month and over three full days per year, thus becoming a chronic problem of personal exposure. Many studies have reported that very

short (<2 h) exposure to traffic particles aggravates respiratory and cardiovascular illnesses, leading to a rise in emergency room visits, as well as the risk of morbidity and mortality in the long term (e.g. Sinharay *et al*., 2018; Laeremans *et al*., 2018; Weichenthal *et al*., 2014; Peters *et al*., 2013; McCreanor *et al*., 2007). Although the health effects associated with each pollutant species are still under discussion, current evidence suggests that actions to reduce air pollution at bus stops will benefit public health and encourage people to use public transport.

The location and configuration of a bus shelter within the streetscape are factors to consider when designing a bus stop to minimize exposure to traffic pollution. However, designs of bus stops respond usually to functional factors, such as commuter's volume and socio-demographic characteristics, accessibility, congestion on buses, interaction with private traffic, distance between bus stops, bus frequency, and operation costs (Ibeas *et al*., 2010), while little or no attention is paid to particle exposure. The orientation of a shelter significantly affects the exposure to particles in its interior. Particle concentration inside a shelter oriented towards the roadway tends to be higher than outside the shelter (Moore *et al*., 2012; Hess *et al*., 2010). Similarly, shelters with poor ventilation, as those enclosed by three sides, may act as particle traps, limiting dispersion and increasing exposure concentration (Moore *et al*., 2012). In the same context, the distance of a bus stop from a street intersection is another factor to consider (Choi *et al*., 2018). Bus stops are typically located at or near intersections to provide commuters easier connection to other buses, and to allow buses to stop and start safely in traffic conditions.

One potential solution to reduce exposure to traffic particles at busy bus stops is to equip shelters with filtering systems. This idea was explored by a group of entrepreneur engineers based in Singapore, who harnessed available technology to design and build an innovative shelter equipped with an advanced air cooling and filtering system (ST Engineering, 2018). A first prototype was erected alongside an existing bus stop. It featured overhead nozzles, that according to the designers, delivered cool air as low as 24°C and filtered out more than 90% of $PM_{2.5}$. The 'Airbitat Oasis Smart Bus Stop' as it was called (Airbitat shelter in short hereafter), was on trial for a period of one year.

To assess the effectiveness of the shelter designed by ST Engineering to mitigate particle exposure in bus stops, the characteristics and concentration of particles in the Airbitat prototype bus shelter were measured and compared to those in a conventional bus stop shelter located right next to it. The hypothesis to be tested was that a significant improvement would not be achieved by equipping open bus shelters with cooling and filtering systems.

When the Airbitat pilot bus stop opened in March 2018, Singapore's Land Transport Authority (LTA) said it was pleased to provide opportunities for local companies to test bed ideas that could make commuting more pleasant (Lim 2018, 2018). This initiative came at the end of the SG$1.1 billion five-year Bus Service Enhancement Programme, in which 1,000 new buses were introduced (LTA, 2017), and at the time of the public consultation exercise for the Land Transport Master Plan 2040 (LTA, 2019). The impact of a prior study on personal exposure at bus stops should also be highlighted (Velasco and Tan, 2016). The public dissemination of its findings through mass media (e.g. Tan, 2016; Channel NewsAsia, 2016) proved beneficial in boosting public awareness and gaining the authorities' attention and traction that the issue required (e.g. Ministry of Transport, 2019).

Methodology

One week after ST Engineering announced the beginning of the trial period for the Airbitat bus stop, particle concentration in the new shelter and the original shelter were simultaneously measured as a means of evaluating the effectiveness of the proposed filtering system. Five set of measurements were conducted during the evening rush hour (16:20-19:20 h) over five days of the second and third week of March 2018.

The Airbitat shelter was erected alongside an existing bus shelter outside a busy underground train station (Dhoby Ghaut MRT station) in the iconic retail boulevard of Orchard Road located in the heart of Singapore's commercial area. The bus stop is located 60 m before a traffic light. Orchard Road is a five-lane single-carriageway with several pedestrian crossings and intersections. The speed limit is 50 km h^{-1}. One lane of the road is dedicated to buses and accommodates 15 bus services. Traffic is heavy with a volume of 2,170 ± 290 cars h^{-1} (geometric mean ± 1 standard deviation), and 144 ± 32 buses h^{-1} stopping to pick up passengers according to onsite manual counts during the sampling period.

This section briefly describes the characteristics and configuration of the instruments, measurements, and data post-processing. Further details are provided in the Supplementary Material (SM). Details of the Airbitat bus stop are also presented below. It is important to note that the designers of the shelter disclosed minimal technical information. Diagrams and specifications of the different components were not made publicly available. Likewise, the process followed for its design and construction is unknown. This issue precludes a thorough discussion of its performance. However, the available information makes it possible to assess the effectiveness of the shelter in filtering traffic particles and reducing the heat to which commuters are exposed.

Airbitat Oasis Smart bus stop

The main components of the Airbitat bus stop are shown in Figure 1. The central component is a 3-step adiabatic cooling system (Reevac™ Deep Cooling) with an integrated electrostatic air filter. The stream of air first passes through a pre-filter where large particles are trapped. The small particles then pass through an electric field and become charged, and are subsequently attracted to and captured by electrically charged plates. Designers claim at least 90% efficiency in $PM_{2.5}$ removal is achieved. As for the cooling system, cold water is firstly generated by flowing water through panels of evaporative medium. The cold water then feeds a heat exchanger. Ambient air passes over it, rapidly lowering its temperature without adding humidity. The cooled air then passes through a second panel of evaporative medium and emerges as a stream of deeply cooled air, with a temperature of 24°C. As specified by the manufacturer, it cools deeper than conventional evaporative coolers without refrigerants or a compressor, while being up to 70% more energy efficient than air conditioners with similar cooling capacity. The clean, cooled air is then channeled through nozzles and air ducts near the ceiling of the bus shelter and directed to commuters waiting at the stop.

The Arbitat bus stop incorporates advanced machine learning and sensors that detect ambient conditions and commuter traffic to adjust operation modes and deliver energy-smart cooling. According to the designers, the sensors at the bus stop detect operating conditions on the ground (e.g. air temperature (T), relative humidity (RH),

commuter flow, etc.) and dynamically adjust the modes in which the air should be cooled. During a period of relative calm, the system turns off and goes into standby mode. The sensors also provide real-time information on T and $PM_{2.5}$ concentrations inside and outside the shelter via a digital panel for traveler information.

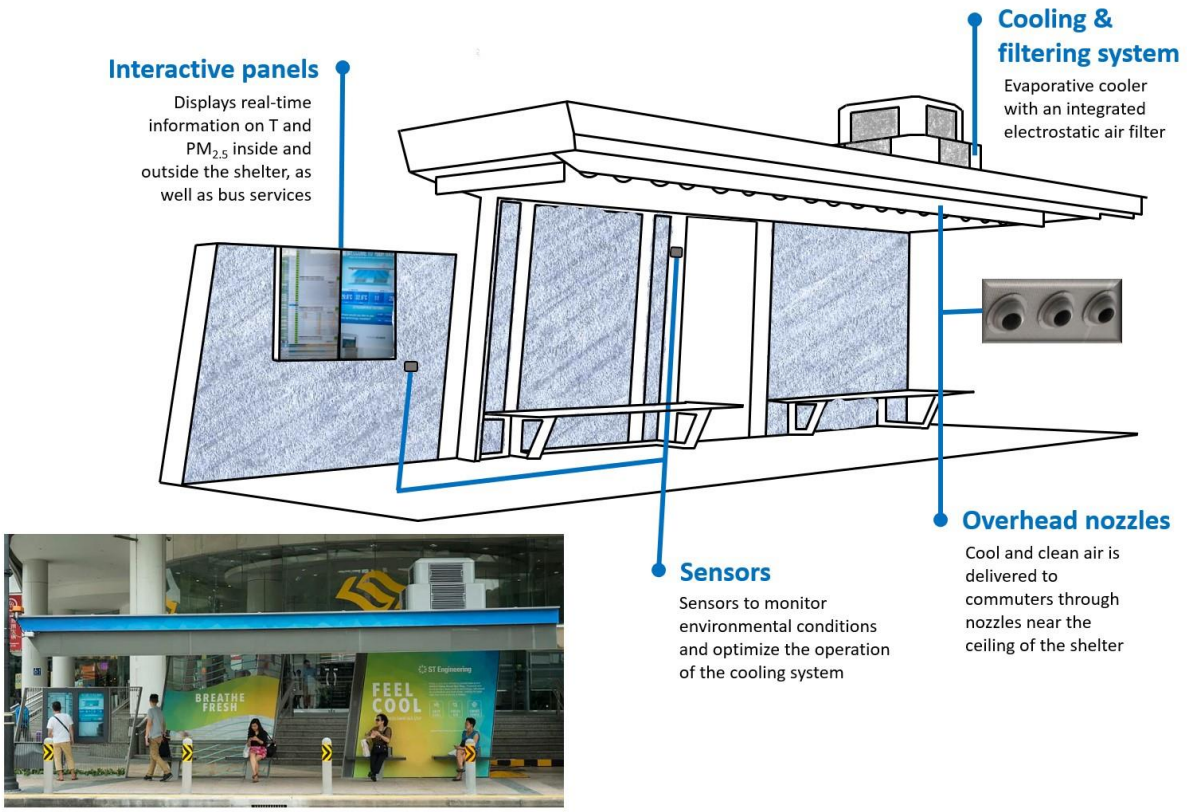


Figure 1. Main components of the Airbitat Oasis Smart Bus Stop, along with a picture showing it in service. An animated explanation of its features is available at https://www.youtube.com/watch?v=-Qpkoqs_ads.

As already mentioned, the technical details for the Airbitat bus stop were not disclosed to the public, but the specifications of the Airbitat Urban Cooler model UBC1 (https://www.airbitat.com) designed to deliver fresh cool air to semi-open spaces of around 30-37 m^2 provide insight on the operating characteristics of the cooling system mounted on the Airbitat bus stop. The Airbitat bus shelter occupied an area of 32 m^2, so the technical features of the Airbitat Urban Cooler model UBC1 could be taken as equivalent to those of the Airbitat bus stop. The datasheet indicates a cooling capacity of 25.5 kW (based on ambient conditions of 34°C and 50% of RH) with an associated air flow of 9,000 $m^3\ h^{-1}$. The cooler consumes 2.9 kW and requires 30 liters of water per hour. Its dimensions are 1.35×1.35×2.20 m and weights 380 and 500 kg, dry and wet, respectively. A running cost of SG$0.77 (US$0.60) per hour was estimated based on tropical climate conditions, and Singapore's utilities calculations.

Instrumentation

Two battery-operated sets of portable sensors were used to simultaneously measure mass concentrations of $PM_{2.5}$ and equivalent black carbon (eBC), and particle number (PN) concentration in both shelters. On alternate sampling days, the mass concentration of

particle-bound polycyclic aromatic hydrocarbons (pPAHs) and the active surface area (ASA) of the aerosols were measured in each shelter, since only one set of sensors to measure these parameters was available. The particle number concentration and ASA were measured as a means to quantify UFP (Heal *et al*., 2012), while eBC was measured as a direct marker of traffic particles (Janssen *et al*., 2011). Noise level, air T and RH were also measured as miscellaneous parameters. Table 1 summarizes the characteristics of the instruments. All sensors were programmed for 1 s readings, with the exception of those measuring pPAHs and ASA, which were programmed for 10 s readings. These frequencies are needed to capture the variability that characterizes particle pollution in microenvironments affected by vehicular traffic according to previous studies (Velasco *et al.* 2019; Tan *et al*., 2017).

Table 1. Instruments used and their measurement characteristics

Parameter	Instrument	Measuring range (lower threshold)	Accuracy[a]	Logging interval (s)	Sampling flow rate (mL min^{-1})	Model	Manufacturer
Size segregated mass-fraction concentration for $PM_{2.5}$	Handheld Dust Trak DRX Aerosol Monitor	0.001-150 mg m^{-3} (1 µg m^{-3})	±0.1% of reading or 0.001 mg m^{-3} [b]	1	3000	TSI 8534	TSI, Shoreview, MN, USA
Particle number concentration (particles < 1 µm diameter)	Handheld Condensation Particle Counter (CPC)	0-1×10^5 # cm^{-3} (1 # cm^{-3})	±20% of reading	1	700	TSI 3007	TSI, Shoreview, MN, USA
Active surface area	Handheld Diffusion Charger (DC)	0-1000 mm^2 m^{-3} (1 mm^2 m^{-3})	±15% of reading or ± 2 mm^2 m^{-3} [b]	10	1000	DC 2000CE	EcoChem Analytics, League City, TX, USA
Equivalent black carbon	Micro-aethelometer	0-1 mg m^{-3} (0.001 µg m^{-3})	±0.1 µg m^{-3}	1	100	AE51	AethLabs, San Francisco, CA, USA
Total pPAHs concentration (particles < 1 µm)	Handheld Photoelectric Aerosol Sensor (PAS)	0-4000 ng m^{-3} (1 ng m^{-3})	±15% of reading or ±3 ng m^{-3} [b]	10	1000	PAS 2000CE	EcoChem Analytics, League City, TX, USA
Noise	Sound Level Meter	30-130 dB (0.1 dB)	±1.5 dB	1	---	CA834	Chauvin Arnoux, Paris, France
Temperature and relative humidity	HOBO ProV2 logger	-40–70 °C, 0-100% (0.01°C, 0.05%)	±0.2°C (0-50 °C) ±2.5% (10-90%) to max ±3.5%	1	NA	U23-001	Onset Computer Corp., Bourne, MA, USA

[a] as reported by the manufacturer
[b] whichever is greater

Each set of instruments was placed on benches in the center of the shelters at 70 cm from the ground. This height is slightly lower than the breathing height of a seated person, but was chosen for practical reasons and for safety of the instruments.

Details on instrument setup and preparation are provided in the SM, as well as a description of the applied corrections and data post-processing. Previous studies have evaluated the responses of the instruments for measuring $PM_{2.5}$, eBC and PN against reference or scientific grade instrumentation, as a means of indirect calibrations. The pairs of instruments, including the sensors for monitoring particles, noise, T and RH, were run side-by-side for 5 min before and after each sampling period at a location near the bus stop that was not directly affected by traffic emissions. The mean differences between pairs of sensors during both testing periods were used to build linear regressions to remove any trends during the measurements and adjust the data sampled in both shelters.

To gain further insight on the origin of the particles, the ratio obtained from the concurrent readings of pPAHs and ASA was used as a fingerprint of the type of combustion particles (Bukowiecki *et al.*, 2002). Following the acronyms of the instruments used to measure both metrics, Photoelectric Charger (PC) for measuring pPAHs and Diffusion Charger (DC) for measuring ASA, the fingerprint is known as PC/DC ratio. It yields distinctive values according to the type of fuel and combustion. For instance, diesel exhaust particles yield ratios of ~1 ng mm^{-2}, while those from gasoline exhaust yield ratios of <0.6 ng mm^{-2} (Ott and Siegmann, 2006). Ratios of >1 ng mm^{-2} are related to high pPAH emissions during periods of hard acceleration due to incomplete combustion (Tang *et al.*, 2001). Particles from non-combustion sources or those that are already coated by condensable species such as semi-volatile hydrocarbons or molecules of water yield a PC/DC ≈ 0.

Similarly, the mean particle size in microenvironments heavily impacted by traffic emissions can be estimated as the diameter of average surface ($D_{Aver,S}$) computed from the simultaneous but independent measurements of PN and ASA (Bukowiecki *et al.*, 2002; Kittelson *et al.*, 2000). Because >90% of the number of particles within exhaust plumes fall in the nucleation mode and are formed as the hot exhaust gases cool and condense after passing the emission control devices, the assumption of spherical particles of this approach can be used as a fair approximation to estimate their average size. $D_{Aver,S}$ represents the diameter of a hypothetical monodisperse particle that has the same ASA as the measured polydisperse particle.

Results

The five set of measurements returned $PM_{2.5}$ concentrations ranging from 32 to 40 µg m^{-3} and 29 to 39 µg m^{-3} (25th and 75th percentiles, hereinafter unless otherwise stated) in the original and Airbitat shelters, respectively. Similarly, PN concentrations of 21,300-37,300 # cm^{-3} and 20,700-36,100 # cm^{-3}, and eBC concentrations of 5-19 µg m^{-3} and 5-20 µg m^{-3} were measured in both shelters. These results come from more than 15 h of sampling, in which around 50-thousand readings were collected for each particle metric.

Although the particle concentrations measured in both shelters were only slightly different, the differences between the pair of medians observed on each sampling day were generally statistically different. The Mood's median test and Kruskal-Wallis test, both performed with a 95% confidence level ($p \leq 0.05$) showed that the variations in $PM_{2.5}$, PN and eBC were significant. Previously, the Anderson-Darling test was applied and it was found that these variables were not normally distributed but positively skewed, hence the use of nonparametric tests to investigate the equality of results obtained between shelters.

Figure 2 shows the differences in $PM_{2.5}$, PN and eBC concentrations, noise levels, T and RH measured simultaneously in both shelters on each sampling day. The difference in the number of commuters waiting at both shelters is also included, although they were not counted simultaneously. The means and variability ranges observed in both shelters for each set of measurements are presented in the SM (Table SM1-SM11 and Figure SM3).

Statistically significant differences were also observed for T, RH and noise levels, except in one set of measurements for the latter parameter. Temperatures of 31.2-33.4°C and 30.9-32.5°C, and RH values of 48.0-60.9% and 52.0-62.3% were measured in the original and Airbitat shelters, respectively. While, noise levels of 72.6-77.9 dBA and 73.1-78.2 dBA were recorded in both shelters throughout the whole study.

Concentrations of pPAHs of 83-343 ng m^{-3} and 54-249 ng m^{-3}, and ASA levels of 164-310 mm^2 m^{-3} and 97-233 mm^2 m^{-3} were observed in the original and Airbitat shelters, respectively. It was not possible to make a comparison between shelters for these two variables because they were not measured simultaneously in both shelters, as already mentioned, the measurements were made on alternate sampling days in each shelter. However, the PC/DC ratio obtained from the slopes of the linear-least-square fits (with the *y*-intercept of the line forced through zero) between the concurrent measurements of pPAHs and ASA exhibited similar values in both shelters: 0.77-0.87 ng mm^{-2} in the original shelter, and 0.65-0.82 in the Airbitat shelter throughout the study. The mean particle size obtained from the $D_{Aver,S}$ was also similar in both shelters, particles between 42 and 52 nm were observed in the original shelter, and between 38 and 48 nm in the Airbitat shelter.

On each measuring day, the number of commuters in the shelters was counted twelve on average. There were between 11 and 19 commuters at any time in the original shelter, and between 7 and 10 in the Airbitat shelter. The number of commuters was statistically different in both shelters in two days of measurements.

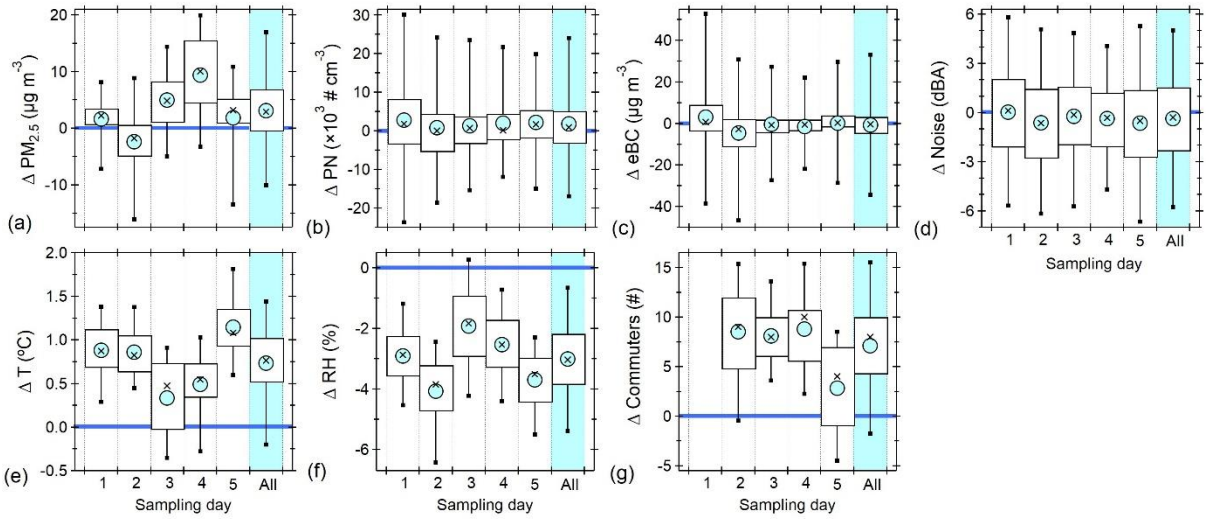


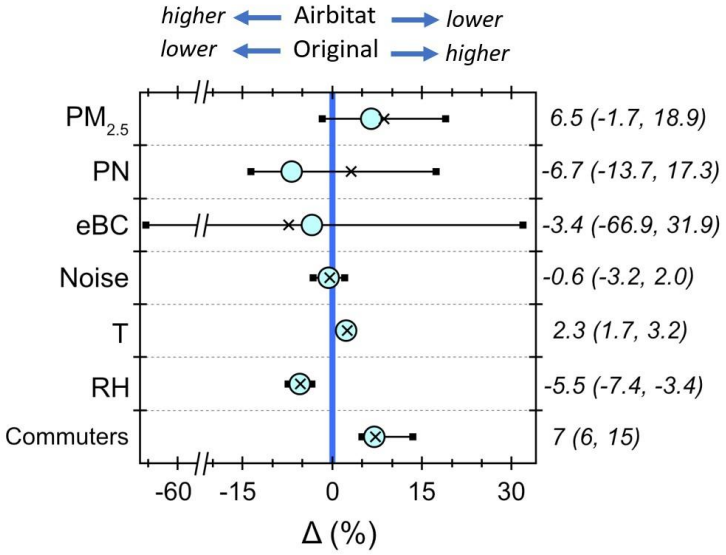
Figure 2. Differences in $PM_{2.5}$, particle number and equivalent black carbon concentrations (a-c), noise level (d), temperature (e), relative humidity (f), and number of commuters (g) between the original and Airbitat bus shelters (i.e. original – Airbitat) for each set of measurements. The box plots on a blue background correspond to the differences considering all the data collected during the five days of measurements. Records above zero (blue thick line) indicate higher concentrations or values at the original shelter, and below zero higher concentrations or values at the Airbitat bus shelter. In each box, the blue dots and crosses are arithmetic means and medians, respectively, while the top and bottom lines are upper and lower quartiles (75th and 25th percentiles). Whiskers extend to the 95th and 5th percentiles.

Discussion

According to the findings, bus shelters equipped with filtering systems based on electrostatic precipitators do marginally reduce the load of $PM_{2.5}$, but not of UFP and eBC to which commuters are exposed while waiting for the bus. The cooling system produced slightly cooler and slightly more humid air, which, when combined with the airflow from the nozzles located near the ceiling, might slightly improve commuters' thermal comfort. The cooling and filtering systems mounted on the Airbitat shelter did not produce additional

noise. Figure 3 shows the percentage difference between both shelters for each parameter that was measured over the course of the five-day study. The reasons why the Airbitat shelter did not achieve a greater reduction in particles load, as well as the differences observed between shelters for the other measured parameters, are discussed below.

Figure 3. Difference in percentage between both shelters (original - Airbitat) for each parameter that was measured throughout the study. Positive differences (i.e. to the right of the blue thick line) indicate higher concentrations or values in the original shelter, while negative differences (i.e. to the left of the blue thick line) indicate higher concentrations or values in the Airbitat bus shelter. Blue dots and crosses indicate arithmetic means and medians, respectively. Whiskers extend to the 75th and 25th percentiles. Figures to the right are means, and 75th and 25th percentiles in parenthesis.



The readings of T and $PM_{2.5}$ concentrations from the sensors installed in the Airbitat shelter were not publicly available, so a comparison against the measurements of this study was not possible. They were displayed for very short periods of time (i.e. a few seconds) on the digital panel mounted into the shelter. In any case, such $PM_{2.5}$ data would be of little help. Examining the shelter's structure revealed that the particle monitors were constructed using low-cost sensors that appeared to be of questionable quality (Libelium Particle Monitor M-X, www.libelium.com). The manufacturer does not provide technical details and it is unclear if the sensors were properly calibrated and adjusted to local conditions as recommended by the World Meteorological Organization (WMO) and the atmospheric scientific community (WMO, 2021).

Effectiveness to reduce particle loads

Four set of measurements recorded higher concentrations of $PM_{2.5}$ in the original shelter (Figure 2a), exceeding each one on average 1.8-9.4 µg m^{-3} the concentrations observed in the Airbitat shelter. Taking into account the full set of simultaneous readings throughout the five-day study, the Airbitat shelter shown to be capable of reducing concentrations by 3.1 µg m^{-3} on average. This represents a reduction of 6.5%, which is insufficient to halve the particles load needed to achieve the hourly mean $PM_{2.5}$ concentration of 16 µg m^{-3} usually reported by the local air quality monitoring network during the evening rush hour across the city (https://www.haze.gov.sg/).

Regarding UFP, no clear difference was found in PN concentrations between shelters (Figure 2b). Although concentrations in the original shelter were on average ~1,000 # cm^{-3} higher than those observed in the Airbitat shelter, the analysis of the full set of simultaneous readings showed that concentrations could be higher by up to 17% (5,140 # cm^{-3}) in the original shelter, but also in the Airbitat shelter by up to 14% (3,310 # cm^{-3}). Similarly, no clear differences were observed for eBC. The differences ranged on average

between zero and 4.7 µg m^{-3} without an evident trend in favor of either of the two shelters (Figure 2c), but with the peculiarity that the instantaneous differences were highly variable (see Figure 3).

Electrostatic precipitators work best with particles larger than 1 µm and generally have lower removal efficiencies than mechanical filters. They are used to remove coarse and fine particles, but not UFP (Liu *et al*., 2017). Therefore, it is not surprising that the filtering system installed in the Airbitat bus shelter failed to significantly reduce the abundance of UFP and eBC. In the best case, electrostatic precipitators like the one discussed here (based on the limited technical details provided by the designer), present efficiencies no better than 90% and 20% for filtering $PM_{2.5}$ and UFP, respectively. It should also be noted that the corona discharge used to ionize the particles passing through the electrostatic precipitator generates ozone (O_3), which is another air pollutant of great concern in urban atmospheres. However, it has been found that by integrating fibrous filters and dielectric plates, the collection efficiency of electrostatic precipitators can increase to levels close to 100% for $PM_{2.5}$ and above 90% for UFP, while reducing O_3 production (Chen *et al*., 2020; Feng and Cao, 2019; Li *et al*., 2015).

It is also important to consider the origin of the particles to understand why the Airbitat shelter reduces the $PM_{2.5}$ load only slightly, and the UFP and eBC loads even less. Recent studies have found that traffic-related emissions have less influence on $PM_{2.5}$ concentrations than on UFP and eBC concentrations at monitoring sites near roads. For example, source apportionment studies in Toronto and Vancouver, Canada revealed that vehicular emissions including exhaust and non-exhaust (i.e. brakes abrasion, tire wear and resuspension of road dust) emissions account for 15-40% of $PM_{2.5}$, but 45-70% of UFP and eBC at road sites (Dabek-Zlotorzynska *et al*., 2019; Hilker *et al*., 2019; Jeong *et al*., 2019). Thus, the $PM_{2.5}$ load varies less inside the shelter as a result of being less dependent on traffic emissions, which makes it easier to filter, unlike UFP and eBC loads, which are more sensitive to fresh emissions from buses idling and speeding up. Statistical analysis showed that $PM_{2.5}$ concentrations varied less than UFP and eBC concentrations (see Table SM1-SM3).

Clean air injected through nozzles installed near the ceiling of the Airbitat shelter had no effect on the particle characteristics. Both shelters registered very similar PC/DC ratios and $D_{Aver,S}$ values. Although the average PC/DC ratio of 0.75 observed in both shelters throughout the study is between those reported for emissions from gasoline and diesel vehicles (Ott and Siegmann, 2006; Velasco *et al*., 2004), individual ratios close or above one were frequently observed. Diesel vehicles typically yield ratios around one and, in this study, such ratios show the strong influence of pPAH-rich fumes emitted by diesel-powered public buses on the particle load. In Singapore, diesel vehicles are mostly goods vehicles and buses, passenger cars run on gasoline, while taxis used to be diesel, but more than half have switched to petrol-electric models in recent years. Similarly, the $D_{Aver,S}$ of 46 ± 10 nm observed in both shelters falls in the upper range of the nucleation mode and in the lower range of the accumulation mode particles, as expected from diesel and gasoline engines equipped with catalytic traps and converters, respectively (Burtscher, 2005; Bukowiecki *et al*., 2002).

Ceiling clean air streams found to be effective at dispersing short-lived UFP spikes associated with fumes from accelerating buses. A commuter can expect one spike above 100×10^3 # cm^{-3} every 10 min in the original shelter, but none in the Airbitat shelter.

Similarly, one spike exceeding 75×10^3 # cm^{-3} can be expected every 5 min in the original shelter, but only one every 13 min in the Airbitat shelter. Regarding $PM_{2.5}$, there was no reduction in the number of spikes in the Airbitat shelter. Every 20 min a spike above 100 µg m^{-3} can be expected in both shelters, and every 10 min a spike over 75 µg m^{-3}.

The orientation and position of a bus stop shelter within the streetscape is another factor to take into account to reduce the particle load to which commuters are exposed. A complete picture of the airflow within the shelter considering the urban canyon is needed to develop designs truly capable of reducing the abundance of particles. Previous studies have shown that poorly ventilated shelters facing the roadway favor the accumulation of particles (Moore *et al*., 2012; Hess *et al*., 2010). The air stream injected by the nozzles mounted on the edge of the ceiling along the open side of the shelter increased circulation, but the flow of air coming in from outside was unknown. This study did not include measurements of wind speed and wind direction inside and outside the shelter, so it was not possible to assess improvements in ventilation. However, a recent study based on computational fluid dynamics (CFD) demonstrated that air curtains installed in the open side of bus shelters reduce the inflow of polluted air, as long as the air is injected toward the road and not toward the shelter, as it was done for the case of the Airbitat shelter (Noh *et al*., 2021). These authors found that the greatest reduction of particles is achieved with an injection angle of 30° with respect to the vertical plane.

Differences in noise level, temperature and relative humidity

The operation of the Airbitat shelter cooling and filtering system did not increase the noise level at the already noisy bus stop. An equivalent continuous sound level (L_{eq}) of 75.6 ± 4.3 dBA was recorded in both shelters. Noise levels over 80 dBA were present 13% of the time, exhibiting one peak over 90 dBA every 10 min, and over 95 dBA once an hour. This noise level classifies as “very annoying” by most scales of reference (e.g. Ouis, 2001), and has a significant impact on people's mood and health. According to Singapore’s legislation on noise pollution (National Environmental Agency (NEA), 2020) and considering scheduled bus intervals of no more than 10 min during peak hours as stipulated by current transport regulations (Ministry of Transport, 2023), the noise level in both shelters exceeded the permissible noise limit (L_{eq} = 75 dBA) in commercial premises for periods no longer than 5 min.

Cross-sectional epidemiological studies have found that long- and short-term exposure to both traffic noise and air pollution have a direct influence on human health (Daiber *et al*., 2022; Stansfeld, 2015). Although a synergistic mechanism between both environmental stressors has yet to be found, their independent impacts have been well documented. Both cause oxidative stress which, as already mentioned, leads to respiratory illnesses, as well as increase the risk of cardiovascular disease, hypertension, and mild cognitive impairment (Fucks *et al*., 2017; Tzivian *et al*., 2016; Hemmingsen et al., 2015; Kälsch *et al*., 2014; Beelen *et al*., 2009). In this context, it is important to safeguard commuters from excessive noise at bus stops.

Despite the fact that the RH increased by 3% on average and the T decreased by 1°C, it is likely that the Airbitat shelter had improved thermal comfort due to the air streams injected through the overhead nozzles. A complete assessment of thermal comfort was not possible because the study did not include measurements of mean radiant temperature and wind speed (Johansson *et al*., 2014). Future studies should consider these variables. Surveys

performed by media sources covering the occasion at the launch of the Airbitat shelter revealed that commuters generally welcomed the cooler wait (e.g. Lim, 2018; Choo, 2018). However, the number of commuters waiting for the bus at both shelters during the study suggested otherwise. On average, 14 commuters were counted at any one time in the original shelter, but only 8 in the Airbitat shelter. According to these numbers, the commuters may have even found the new shelter uncomfortable rather than noticing a thermal improvement. In hot environments, the loss of metabolic heat is dominated by evaporation, and increased humidity exacerbates thermal discomfort (Nicol, 2004). Also keep in mind that the body reacts slowly to temperature changes, and commuters might not have enough time to notice improvements in thermal comfort when waiting at a bus stop.

Conclusion

The Airbitat bus shelter was unable to significantly reduce the particle load and heat to which commuters are exposed to while waiting for the bus, contrary to what was originally claimed by its designers. Equipping bus shelters with cooling and filtering systems might not be the best solution to mitigate particle exposure and improve thermal comfort at bus stops. Harnessing available technology to solve the problem is not enough, it must also be adequately described and fully understood. Advice from fluid dynamics experts on the airflow patterns inside and outside the shelter would have assisted in determining the optimal angle to inject air streams from the shelter's ceiling to better block traffic particles from entering. Similarly, an air pollution specialist would have advised on the need of focusing on ultrafine particles and black carbon rather than $PM_{2.5}$ to design the filtering system, as well as warned about the potential generation of ozone when using an electrostatic precipitator.

Future designs of bus shelter, such as the Airbitat Oasis Smart Bus Stop, must be based on validated information. Before building a full-scale physical prototype, it is strongly recommended to first create virtual prototypes and evaluate their performance using numerical simulations (e.g. CFD). If a wind tunnel is available, smaller-scale prototypes can also be built to assess the effectiveness of proposed designs through fluid dynamics experiments.

It is critical to consider people's needs and preferences while planning urban infrastructure. The socio-psychological aspects are equally important as the technical aspects. A social scientist, for example, may help enhance future designs by clarifying why many commuters did not chose to stand at the Airbitat shelter.

Smart and sustainable cities are the outcome of entrepreneurial and daring ideas developed by engineers, architects, and urban planners who always back their work with solid evidence. The success of an engineering project like the Airbitat bus shelter depends on applying the scientific method from the start. The project must be driven by the desire to solve a well-defined and understood problem, else failure is likely.

Contrary to what was promised at its launch, the Airbitat shelter was dismantled after a year-long trial without disclosing information about its performance or telling anyone what happened to it. The original idea was to retrofit the experience to other existing bus stops, making the concept easily transferable.

Failure of a prototype should not be seen negatively; rather, it should be used as an opportunity to learn and develop a new prototype with updated specifications that achieves the intended outcomes. The Airbitat Oasis Smart bus stop should be seen as the first step toward developing a modern bus stop shelter that lessens commuters' exposure to traffic pollutants and enhances their thermal comfort while they wait for the bus.

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